



Department of Computer Science and Engineering

Academic Year 2024– 2025(Odd Semester)

Degree, Semester & Branch: B.E III & ECE ‘B’

Course Code & Title: CS3353 & C Programming and Data Structures

Name of the Faculty member (s): Mrs.K.Priya, AP/CSE

Innovative Practice Description

- **Unit / Topic: Unit IV / Hashing- Hash Functions**
- **Course Outcome: CO4**
- **Topic Learning Outcome: TLO11**

Activity Chosen: MindMap

- **Justification:**

Mind maps are one of the most common diagramming tools that help students learn new things about any given topic. student can use to demonstrate what they have learned in a visually creative manner. It allows them to explore their knowledge, apply critical thinking skills and summarise key concepts on a page.

- **Time Allotted for the Activity: 15 Minutes**

- **Details of the Implementation:**

- Faculty explained the concept of Hashing and Hash Functions.
- Based on the discussion the teacher asked the Student to draw a mind map related to the topic within 15 minutes.
- The structure of the mindmap involves the following layer concepts
Central Node : The central node represents the Hashing and Hash Functions. It connects all the layers and sub-topics, showcasing how hashing and hash functions integrate into various applications
First Layer : This layer involves the concepts of Hashing and Hash Functions.
Second Layer: This layer focuses on characteristics of Hashing.
Third Layer : This layer demonstrates how hashing is used in real-world scenarios.
Fourth Layer : This layer focuses on emerging and creative uses of hash functions.
Fifth Layer : This layer focuses on Challenges.
- Each student created a mind map on Hashing- Hash Functions.
- The faculty collected the sheet and appreciated the students.
- When comparing student mind maps for hashing and hash functions, the approach will vary based on the level of understanding, focus areas, and depth of exploration into data structures that use hashing.
- A beginner student will focus on the fundamental concepts and applications of hashing in simple data structures such as hash tables.
- An intermediate student would go further into the concept of hashing in data structures and may start exploring variations of hash tables and collision resolution techniques.



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

- An advanced student would deeper into more sophisticated uses of hashing in complex data structures and how hash functions optimize performance.
- Some Students focuses on practical applications of hashing and hash functions in real-world data structures, likely with an industry or system design perspective.

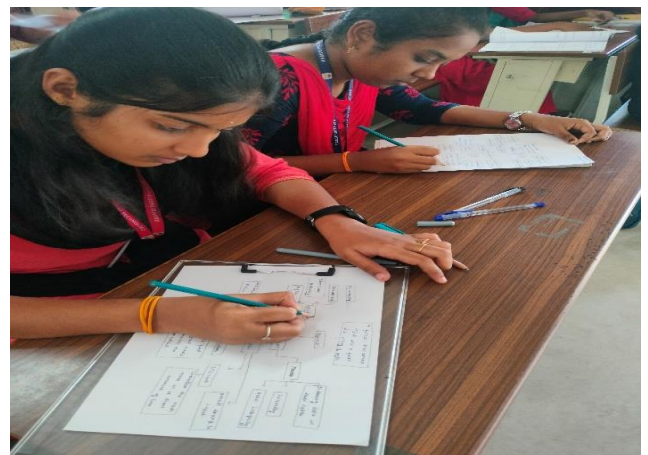
CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12	PSO1
CO4	3	3	2	2	3	1	2	1	1

(1 – Low 2 – Moderate 3 – High)

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12	PSO1
Justification for correlation	Students will be able to apply non-linear data structure concepts to identify solution to complex engineering problems.	Students will be able to Identify, and analyse engineering problems with hashing and tree data structures concept.	Students will be able to design solutions for complex engineering problems using different data structures operations.	Students will be able to provide complex problem solutions with relevant data structure concepts and operations.	Students will be able to use Visualgo.net tool for performing various operations of hashing and non-linear data structure such as trees.	Students will be able to write program individually to perform different operations of hashing and tree non-linear data structure.	Students will be able to explain and present different operations of hashing and tree non-linear data structure.	Student will become aware of the need for lifelong learning and the continued upgrading of technical knowledge using the concept of non-linear data structure.	Students will be able to apply hashing and non-linear data structure concept to solve various application in Information and Communication Technology.

Images or Screenshot of the Practice



Hash Hashfunction:

- Definition:** A function that converts an input into a fixed-size string of bytes.
- Purpose:** Indexing data in hash tables.
- Security:** Data integrity.
- Applications:**
 - Blockchain technology
 - Digital signatures
- Characteristics:**
 - Deterministic:** Same input will always produce the same output.
 - Collision resistance:** Difficult to find two different inputs that produce the same output.
 - Efficient:** Computes the hash value in a short amount of time.
 - Small changes to input:** Significantly alter the output.

Hashing and Hash Function.

- Definition:** A function that converts an input into fixed-size string of bytes.
- Purpose:** Indexing data in hash tables.
- Security:** Data integrity.
- Applications:**
 - Blockchain technology
 - Digital signatures
- Characteristics:**
 - Deterministic:** Same input will always produce the same output.
 - Collision resistance:** Difficult to find two different inputs that produce the same output.
 - Efficient:** Computes the hash value in a short amount of time.
 - Small change in input:** Significantly alter the output.
- Pre-image resistance:** Infeasible to reverse engineer the original input from the hash output.

Reflective Critique:

- Students found the mind map effective for visualizing how different elements of hashing are interconnected, which simplified their understanding of collision handling, applications, and specific algorithms.
- However, a few students mentioned that they would benefit from a hands-on coding example or real-world scenario alongside the mind map to deepen practical understanding.

- Students will be able to build chapter outlines
- Students will identify and analyse primary issues in group work
- Students prepare a chart for Dream Jobs
- Students can make plans with different job interviews
- Students can plan vacations & organizing events.



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

❖ *Challenges faced in implementation:*

- The mind map format focused heavily on theory, making it hard for students to see practical applications.
- There was limited time to cover all mind map topics in detail.
- A mind map is often an individual exercise, but students may feel they benefit more from group discussions where they can clarify doubts with peers and collaboratively build out the map.
- Hashing includes both foundational and applied concepts (e.g., theoretical properties vs. practical applications), which might make it hard for students to understand the “big picture” and how each part connects.

References:

<https://www.ritrjpm.ac.in/images/computer-science/5CS8591Mindmap.pdf>
<https://www.lucidchart.com/pages/how-to-make-a-mind-map>
<https://www.ritrjpm.ac.in/images/computer-science/Mind%20Map.pdf>

Signature of Faculty Member

HOD-CSE